Update on 2025 and 2026 tracking efficiencies

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4th June 2026

Tag-and-probe method

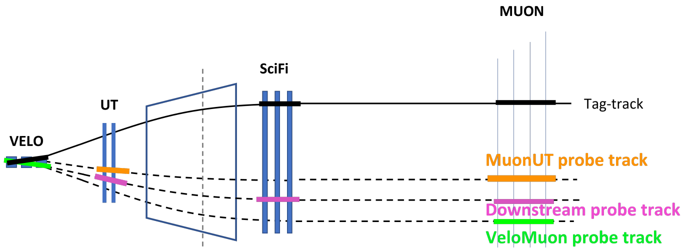


Figure by Rowina

- Tag-and-probe method with $J/\psi \to \mu^+\mu^-$
- Fully reconstruct one muon...
- ... and partially reconstruct the other muon
- Match hits in specific sub-detector with partially reconstructed track

$$\epsilon_{\text{track}} = \frac{N_{\text{matched}}}{N_{\text{matched}} + N_{\text{failed}}}$$

Introduction to methods

Strategy for tracking efficiency determination:

- Combine efficiencies of two methods:
  - VeloMuon: SciFi efficiency
  - Downstream: Velo efficiency
  - Combined: Velo $\times$ SciFi efficiencies
- Cross check with long track efficiency from the MuonUT method
  - Main systematic

Overview of work

Recent presentations:

- A/S week
  - 2024 v0 tables available on GitLab here
- RTA WP4 talk on muon tracking efficiencies
- RTA WP4 talk on hadronic corrections

Tracking efficiencies are determined using TrackCalib2

- GitLab repository here
- Mass fit done on tuples from AP to extract efficiencies
- Note: Analysts should not have to run TrackCalib2 themselves, unless the released numbers are not suitable (need different binning, etc)

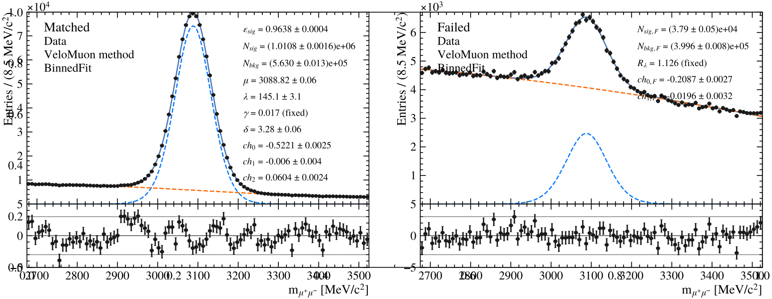
Datasets considered

Datasets considered:

- 2024
  - v0 results for blocks 2, 1, 5, 6, 7, 8, *pp* ref available
  - Currently waiting for large MC production (16M per block) to finish
  - Blocks 4 and 3 don't have MC yet
- 2025
  - Today: Analysis of pre-TS data
  - Post-TS MC in production (4M per block)
  - Very small (100k) post-TS MC samples available for initial checks
- 2026
  - Today: First look at data
  - Realistic MC not available yet

Example mass fit

Example of fit to extract efficiency (2024 block 5, 1D η bin 3)



- Efficiency is a free parameter in the fit
- Matched and failed samples simultaneously fitted to ensure correct uncertainties

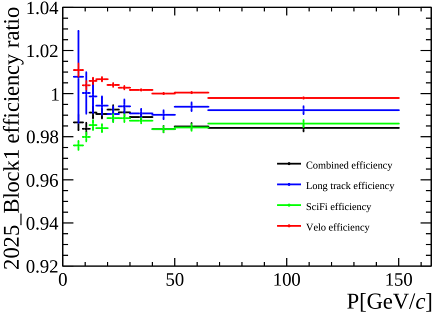
2025 block definitions

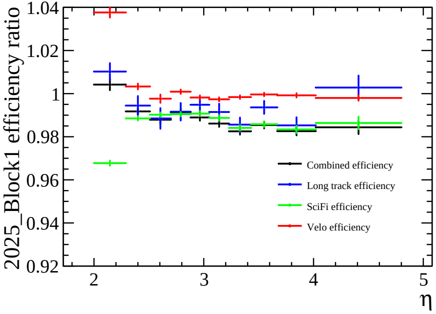
Define 2025 blocks according to simulation conditions:

Block	Sprucing campaign	Magnet polarity	MC sample	$\mathcal{L}_{\text{int}}(\text{fb}^{-1})$
1	Sprucing25c1	MagDown	W21	0.4
2	Sprucing25c1	MagUp	W22.25	1.6
3	Sprucing25c2	MagDown	W28.36	4.3
4	Sprucing25c3/4	MagUp	W37.43	4.1
5	Sprucing5c4	MagDown	W43.45	1.0

2025 data/MC ratios in 1D

2025 pre-TS data/MC ratios, Block 1

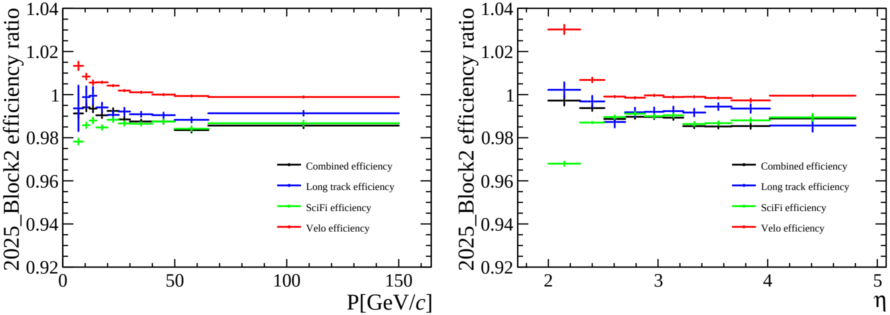




Note that Velo efficiencies are underestimated for this block!

2025 data/MC ratios in 1D

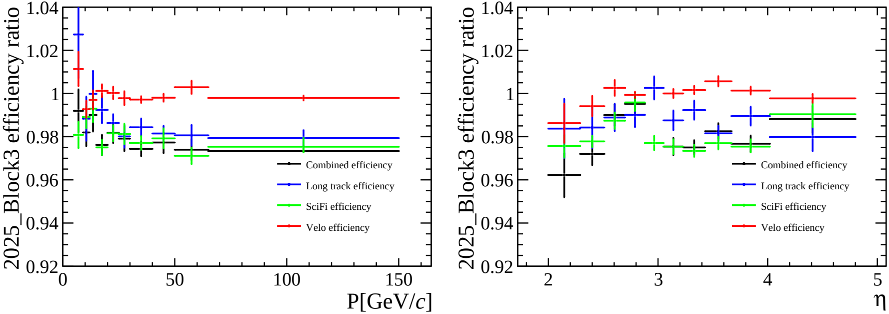
2025 pre-TS data/MC ratios, Block 2



Note that Velo efficiencies are underestimated for this block!

2025 data/MC ratios in 1D

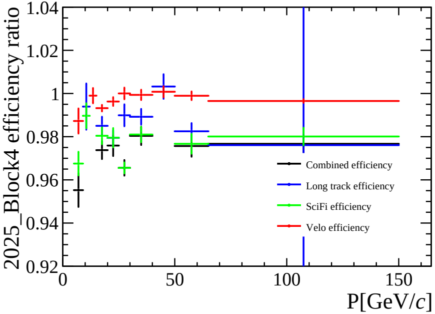
2025 post-TS data/MC ratios, Block 3

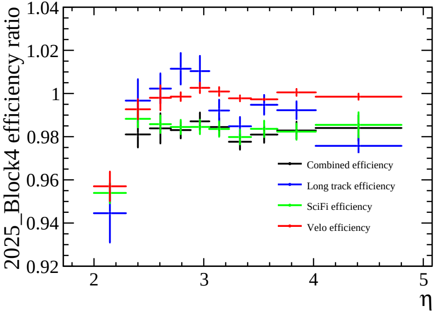


Better Velo efficiency agreement, but MC statistics very low...

2025 data/MC ratios in 1D

2025 post-TS data/MC ratios, Block 4

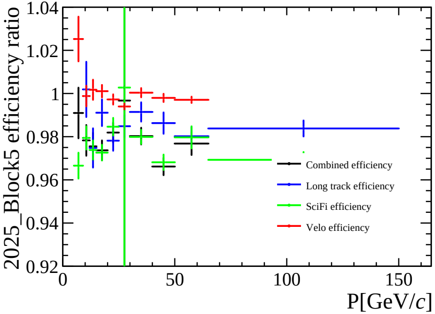


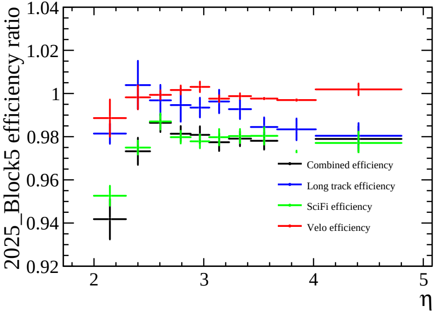


Better Velo efficiency agreement, but MC statistics very low...

2025 data/MC ratios in 1D

2025 post-TS data/MC ratios, Block 5



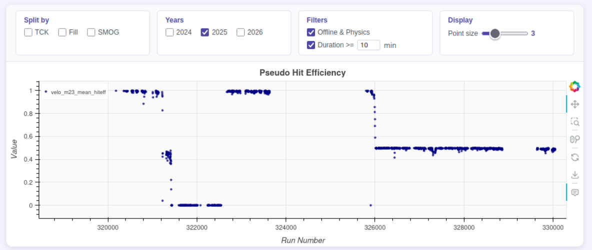


Better Velo efficiency agreement, but MC statistics very low...

2025 pre-TS Velo discrepancies

Why are Velo efficiencies underestimated in pre-TS MC?

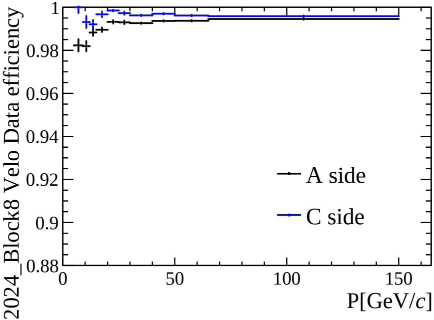
- Velo M23 was (sometimes) working in data, but all four sensors are turned off in MC
- Two sensors in Velo M21 were turned off
- Two consecutive modules that are not working properly can significantly impact track efficiency!

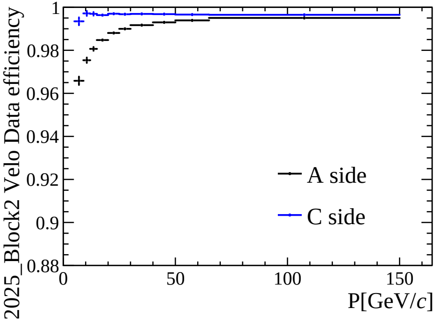


M23 hit efficiency during data taking from Intelli RTA

2025 pre-TS Velo discrepancies

Note: Both M21 and M23 are on the A-side, leading to a significant left-right asymmetry





Only M23 was off in 2024, but in 2025 two sensors failed in M21

2025 pre-TS Velo discrepancies

What are the consequences for tracking efficiencies?

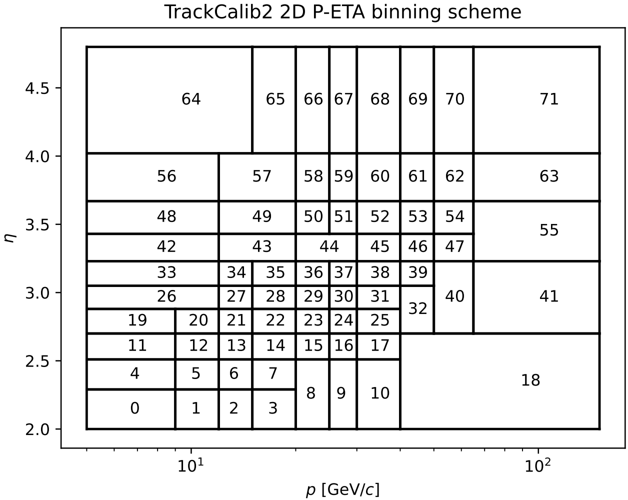
- All Sim10g MC samples are affected
- In principle tracking efficiency corrections should account for this...
- ... but a fix is also in progress: !306

What we would like to do about this:

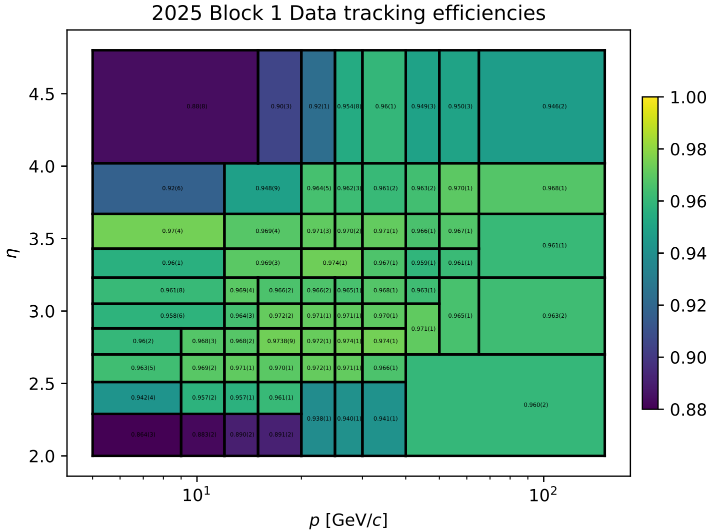
1. Request a larger MC production (16M per block) for 2025 pre-TS, with !306
2. Change p–η binning to p–η–ϕ with two ϕ bins corresponding to left and right ($|\phi| < \pi/2$ and $|\phi| \geq \pi/2$)

2D binning

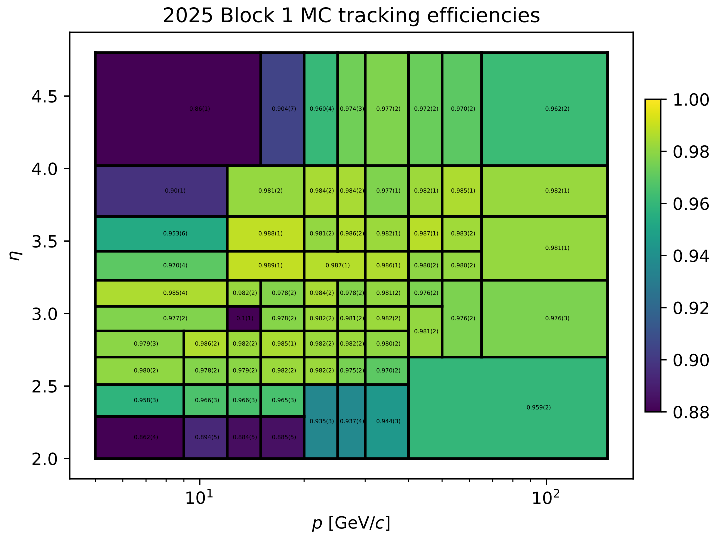
2D binning in p vs η with 72 bins:



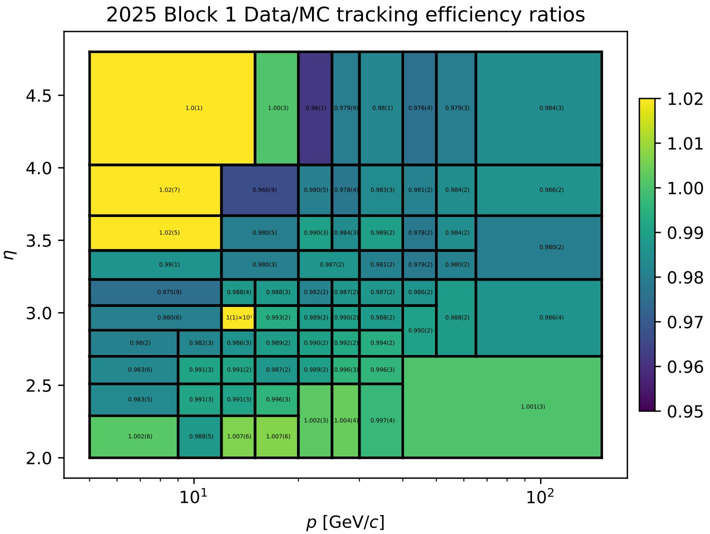
2D binning


2025 Block 1 Data tracking efficiencies
η
4.5
4.0
3.5
3.0
2.5
2.0
10^1
10^2
p [GeV/c]
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.88(8)
0.90(3)
0.92(1)
0.954(8)
0.96(1)
0.949(3)
0.950(3)
0.946(2)
0.92(6)
0.948(9)
0.964(5)
0.962(3)
0.961(2)
0.963(2)
0.970(1)
0.968(1)
0.97(4)
0.969(4)
0.971(3)
0.970(2)
0.971(1)
0.966(1)
0.967(1)
0.961(1)
0.96(1)
0.969(3)
0.974(1)
0.967(1)
0.959(1)
0.961(1)
0.961(8)
0.969(4)
0.966(2)
0.966(2)
0.965(1)
0.968(1)
0.963(1)
0.958(6)
0.964(3)
0.972(2)
0.971(1)
0.971(1)
0.970(1)
0.971(1)
0.965(1)
0.963(2)
0.96(2)
0.968(3)
0.968(2)
0.9738(9)
0.972(1)
0.974(1)
0.974(1)
0.963(5)
0.969(2)
0.971(1)
0.970(1)
0.972(1)
0.971(1)
0.966(1)
0.942(4)
0.957(2)
0.957(1)
0.961(1)
0.938(1)
0.940(1)
0.941(1)
0.960(2)
0.864(3)
0.883(2)
0.890(2)
0.891(2)

2D binning

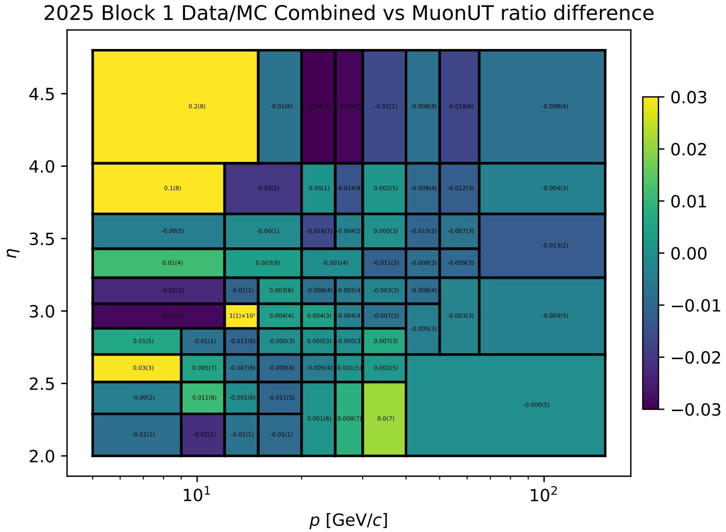

2025 Block 1 MC tracking efficiencies
η
p [GeV/c]
10^1
10^2
2.0
2.5
3.0
3.5
4.0
4.5
0.88
0.90
0.92
0.94
0.96
0.98
1.00
0.86(1) 0.904(7) 0.960(4) 0.974(3) 0.977(2) 0.972(2) 0.970(2) 0.962(2)
0.90(1) 0.981(2) 0.984(2) 0.984(2) 0.977(1) 0.982(1) 0.985(1) 0.982(1)
0.953(6) 0.988(1) 0.981(2) 0.986(2) 0.982(1) 0.987(1) 0.983(2) 0.981(1)
0.970(4) 0.989(1) 0.987(1) 0.986(1) 0.980(2) 0.980(2)
0.985(4) 0.982(2) 0.978(2) 0.984(2) 0.978(2) 0.981(2) 0.976(2)
0.977(2) 0.978(2) 0.982(2) 0.981(2) 0.982(2) 0.981(2) 0.976(2) 0.976(3)
0.979(3) 0.986(2) 0.982(2) 0.985(1) 0.982(2) 0.982(2) 0.980(2)
0.980(2) 0.978(2) 0.979(2) 0.982(2) 0.982(2) 0.975(2) 0.970(2)
0.958(3) 0.966(3) 0.966(3) 0.965(3) 0.935(3) 0.937(4) 0.944(3) 0.959(2)
0.862(4) 0.894(5) 0.884(5) 0.885(5)

2D binning



2025 Block 1 Data/MC tracking efficiency ratios
1.0(1)
1.00(3)
0.96(1)
0.979(9)
0.98(1)
0.976(4)
0.979(3)
0.984(3)
1.02(7)
0.966(9)
0.980(5)
0.978(4)
0.983(3)
0.981(2)
0.984(2)
0.986(2)
1.02(5)
0.980(5)
0.990(3)
0.984(3)
0.989(2)
0.978(2)
0.984(2)
0.980(2)
0.99(1)
0.980(3)
0.987(2)
0.981(2)
0.979(2)
0.980(2)
0.975(9)
0.988(4)
0.988(3)
0.982(2)
0.987(2)
0.987(2)
0.986(2)
0.980(6)
1(1)×10¹
0.993(2)
0.989(2)
0.990(2)
0.988(2)
0.988(2)
0.986(4)
0.990(2)
0.98(2)
0.982(3)
0.986(3)
0.989(2)
0.990(2)
0.992(2)
0.994(2)
0.983(6)
0.991(3)
0.991(2)
0.987(2)
0.989(2)
0.996(3)
0.996(3)
0.983(5)
0.991(3)
0.991(3)
0.996(3)
1.001(3)
1.002(3)
1.004(4)
0.997(4)
1.002(6)
0.988(5)
1.007(6)
1.007(6)
η
4.5
4.0
3.5
3.0
2.5
2.0
10^1
10^2
p [GeV/c]
1.02
1.01
1.00
0.99
0.98
0.97
0.96
0.95

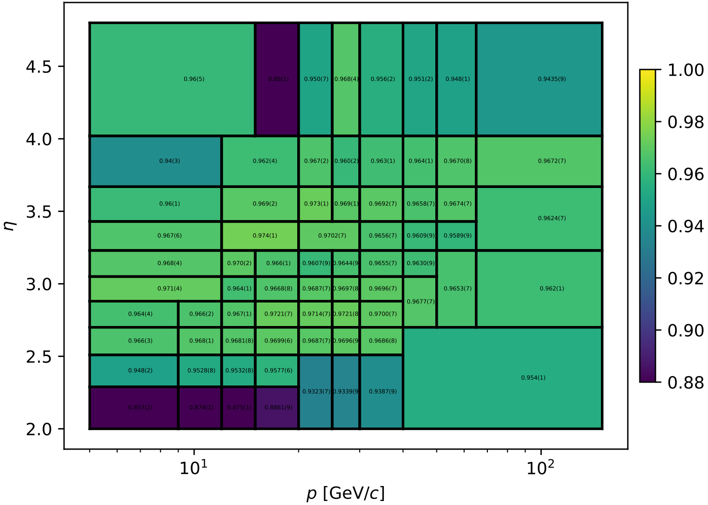
2D binning



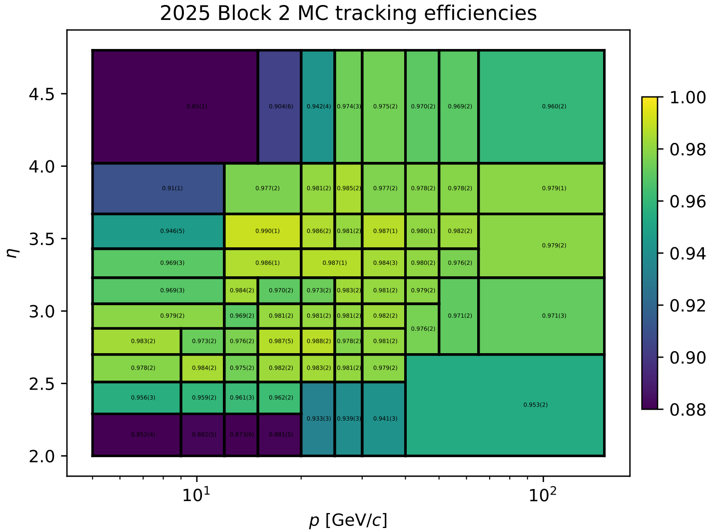
2025 Block 1 Data/MC Combined vs MuonUT ratio difference
η
p [GeV/c]
10¹
10²
2.0
2.5
3.0
3.5
4.0
4.5
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
0.2(8)
-0.01(6)
-0.06(3)
-0.03(2)
-0.02(1)
-0.008(8)
-0.018(6)
-0.008(4)
0.1(8)
-0.02(2)
0.00(1)
-0.014(9
0.002(5)
-0.009(4)
-0.012(3)
-0.004(3)
-0.00(5)
-0.00(1)
-0.016(7)
-0.004(5
0.000(3)
-0.010(2)
-0.007(3)
-0.013(2)
0.01(4)
0.003(9)
-0.001(4)
-0.011(3)
-0.008(3)
-0.009(3)
-0.02(3)
-0.01(1)
0.003(6)
-0.008(4)
-0.005(4
-0.003(3)
-0.008(4)
-0.03(2)
1(1)×10¹
0.004(4)
0.004(3)
-0.004(4
-0.007(3)
-0.005(3)
-0.003(3)
-0.004(5)
0.01(5)
-0.01(1)
-0.011(6)
-0.000(3)
0.000(3)
-0.000(3
0.007(3)
0.03(3)
0.005(7)
-0.007(6)
-0.009(4)
-0.005(4)
0.001(5)
0.002(5)
-0.00(2)
0.011(6)
-0.001(6)
-0.011(5)
0.001(6)
0.008(7)
0.0(7)
-0.000(5)
-0.01(1)
-0.02(1)
-0.01(1)
-0.01(1)

2D binning

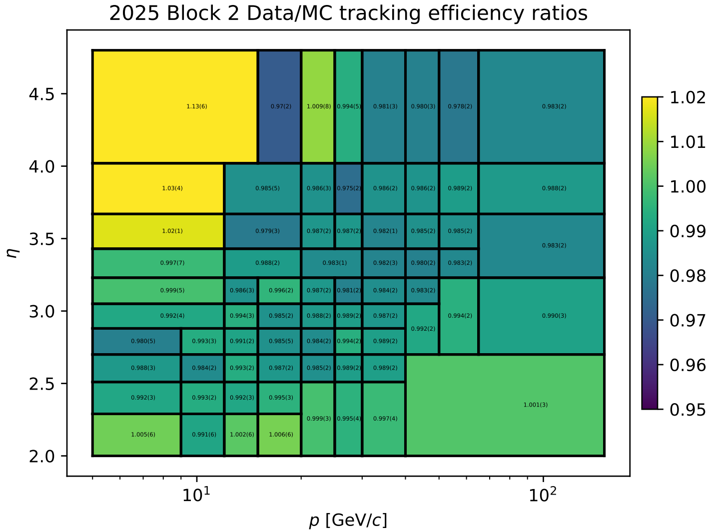
2025 Block 2 Data tracking efficiencies



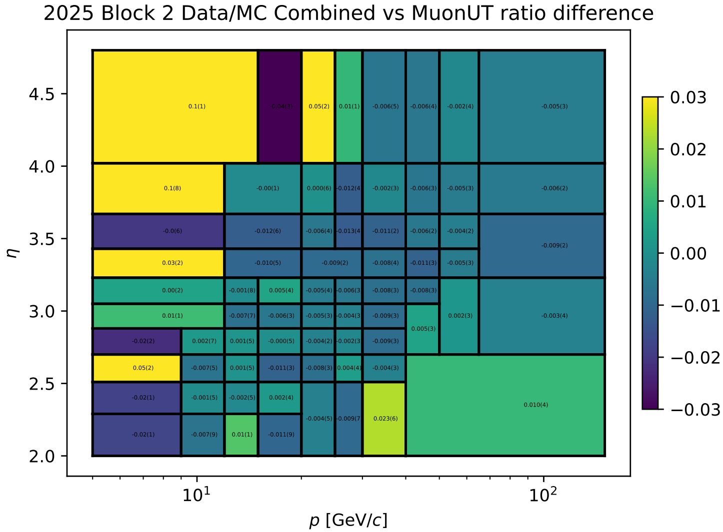
2D binning


2025 Block 2 MC tracking efficiencies
0.85(1)
0.904(6)
0.942(4)
0.974(3)
0.975(2)
0.970(2)
0.969(2)
0.960(2)
0.91(1)
0.977(2)
0.981(2)
0.985(2)
0.977(2)
0.978(2)
0.978(2)
0.979(1)
0.946(5)
0.990(1)
0.986(2)
0.981(2)
0.987(1)
0.980(1)
0.982(2)
0.979(2)
0.969(3)
0.986(1)
0.987(1)
0.984(3)
0.980(2)
0.976(2)
0.969(3)
0.984(2)
0.970(2)
0.973(2)
0.983(2)
0.981(2)
0.979(2)
0.979(2)
0.969(2)
0.981(2)
0.981(2)
0.981(2)
0.982(2)
0.971(2)
0.971(3)
0.976(2)
0.983(2)
0.973(2)
0.976(2)
0.987(5)
0.988(2)
0.978(2)
0.981(2)
0.978(2)
0.984(2)
0.975(2)
0.982(2)
0.983(2)
0.981(2)
0.979(2)
0.956(3)
0.959(2)
0.961(3)
0.962(2)
0.953(2)
0.933(3)
0.939(3)
0.941(3)
0.852(4)
0.882(5)
0.873(6)
0.881(5)
4.5
4.0
3.5
3.0
2.5
2.0
η
10^1
10^2
p [GeV/c]
1.00
0.98
0.96
0.94
0.92
0.90
0.88

2D binning


2025 Block 2 Data/MC tracking efficiency ratios
η
4.5
4.0
3.5
3.0
2.5
2.0
p [GeV/c]
10^1
10^2
1.02
1.01
1.00
0.99
0.98
0.97
0.96
0.95
1.13(6)
0.97(2)
1.009(8)
0.994(5)
0.981(3)
0.980(3)
0.978(2)
0.983(2)
1.03(4)
0.985(5)
0.986(3)
0.975(2)
0.986(2)
0.986(2)
0.989(2)
0.988(2)
1.02(1)
0.979(3)
0.987(2)
0.987(2)
0.982(1)
0.985(2)
0.985(2)
0.983(2)
0.997(7)
0.988(2)
0.983(1)
0.982(3)
0.980(2)
0.983(2)
0.999(5)
0.986(3)
0.996(2)
0.987(2)
0.981(2)
0.984(2)
0.983(2)
0.992(4)
0.994(3)
0.985(2)
0.988(2)
0.989(2)
0.987(2)
0.992(2)
0.994(2)
0.990(3)
0.980(5)
0.993(3)
0.991(2)
0.985(5)
0.984(2)
0.994(2)
0.989(2)
0.988(3)
0.984(2)
0.993(2)
0.987(2)
0.985(2)
0.989(2)
0.989(2)
1.001(3)
0.992(3)
0.993(2)
0.992(3)
0.995(3)
0.999(3)
0.995(4)
0.997(4)
1.005(6)
0.991(6)
1.002(6)
1.006(6)

2D binning



2025 Block 2 Data/MC Combined vs MuonUT ratio difference
η
p [GeV/c]
4.5
4.0
3.5
3.0
2.5
2.0
10^1
10^2
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
0.1(1)
-0.04(3)
0.05(2)
0.01(1)
-0.006(5)
-0.006(4)
-0.002(4)
-0.005(3)
0.1(8)
-0.00(1)
0.000(6)
-0.012(4
-0.002(3)
-0.006(3)
-0.005(3)
-0.006(2)
-0.0(6)
-0.012(6)
-0.006(4)
-0.013(4
-0.011(2)
-0.006(2)
-0.004(2)
-0.009(2)
0.03(2)
-0.010(5)
-0.009(2)
-0.008(4)
-0.011(3)
-0.005(3)
0.00(2)
-0.001(8)
0.005(4)
-0.005(4)
-0.006(3
-0.008(3)
-0.008(3)
0.01(1)
-0.007(7)
-0.006(3)
-0.005(3)
-0.004(3
-0.009(3)
0.005(3)
0.002(3)
-0.003(4)
-0.02(2)
0.002(7)
0.001(5)
-0.000(5)
-0.004(2)
-0.002(3
-0.009(3)
0.05(2)
-0.007(5)
0.001(5)
-0.011(3)
-0.008(3)
0.004(4)
-0.004(3)
-0.02(1)
-0.001(5)
-0.002(5)
0.002(4)
-0.004(5)
-0.009(7
0.023(6)
0.010(4)
-0.02(1)
-0.007(9)
0.01(1)
-0.011(9)

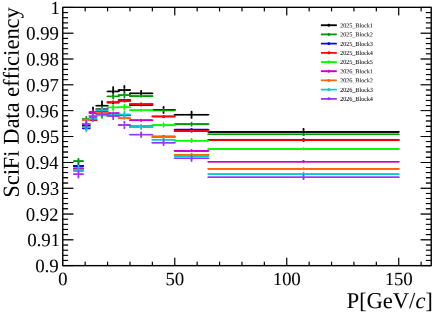
Release of tracking efficiencies

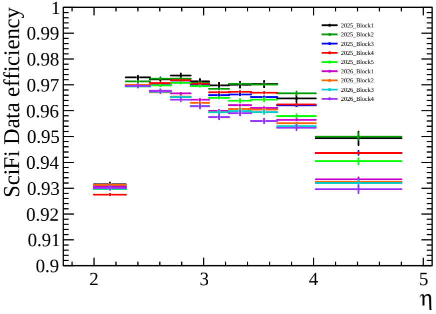
We would like to release the tracking efficiency correction tables for 2025 pre-TS

- 2024 v0 is already available on GitLab
- Add a file for 2025 with the same binning and same format
- Feedback is welcome!

Initial look at 2026 efficiencies

Data SciFi efficiency comparison between 2025 and 2026

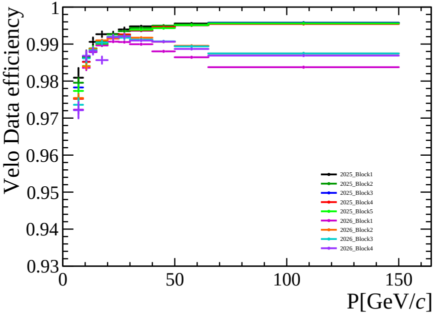


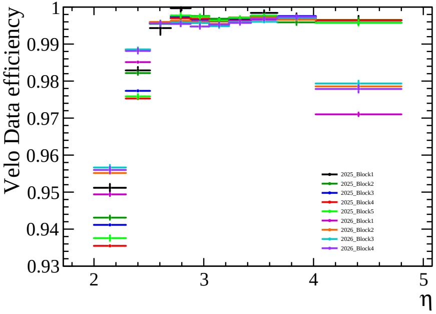


- Drop in efficiencies over time, likely due to radiation damage
- Effect greater in region close to the beam (high η)

Initial look at 2026 efficiencies

Data Velo efficiency comparison between 2025 and 2026





- Clear drop in efficiencies in 2026 at high p and in highest η bin
- Most likely due to damage suffered during 2025 heavy ion run

Summary and next steps

Summary:

1. Behaviour of 2025 tracking efficiencies understood
2. Correction tables for pre-TS have been prepared
3. Post-TS tables will follow once MC is ready
4. See clear drop in Velo efficiencies in 2026 due to damage

Next steps

- Release 2025 post-TS tables (and 2026) once MC is available
- Finalise hadronic corrections
- Update tables using larger MC samples
    - Revised P-ETA binning
    - Separate tables for μ^+ and μ^-
    - Charge integrated table with left and right bins in ϕ

Thanks for your attention!